

# Kaggriculture, Explained to Grandpa

*The Handover — everything you need to run the farm office*

Grab a cup of coffee, boss. You're retiring, and I'm taking over the farm office. But before I sit in your chair, I need to walk you through exactly how this business runs. It's a 30-day season, 24 hours a day — 720 hours in total.

At the end of hour 720, the bank auditor walks in and counts **only the cash in the register**. A field full of ripe melons, a shed full of milk, a worker holding six carrots — the bank values it all at exactly zero dollars. If you didn't sell it and put the money in the register before the season ends, it doesn't count. Everything I'm about to tell you serves that one rule.

## 1. The Walkie-Talkie

You sit in the office with a map and a walkie-talkie. The walkie-talkie is incredibly basic. It only understands short, specific phrases in a strict order. If you use fancy sentences, the men hear static and stand there all day.

You cannot say, “Bob, go to coordinate 3,4 and water the pumpkins.” You say [ "NORTH" ]. Next hour, [ "EAST" ]. And here's the part everyone gets wrong: **Bob can't work the pumpkins from the edge of the patch**. He has to walk right into the patch — boots in the dirt, standing on the plant itself — before you can say [ "WATER" ].

The only exception is the main shed. To drop off a load he just needs to stand on one of the four centre squares touching it: (4,4), (5,4), (4,5), (5,5).

### One slow answer ends the season

You must answer the walkie-talkie the moment it crackles — within **1 second**. This isn't a special rule of this farm; it's how every Kaggle episode works, for any game: a turn that times out, errors, or returns something illegal gets marked TIMEOUT, ERROR or INVALID, and the engine sets that farm's score to nothing at all for the season, not a low score. Same for shouting gibberish down the line, or the office phone dying mid-call.

There is one small mercy: the office keeps a piggy bank of borrowed seconds (`remainingOverageTime`, 60 to start). If you're slow once, the extra time comes out of the piggy bank and you live. But it never refills. Once it's empty, the next slow answer ends you.

On that 60: it isn't a per-competition dial — it doesn't appear in the engine's `configuration` list at all (season length, board size and the rest do), only as a fixed default in the observation schema, applied the same way for every game on this platform. A host can't quietly hand this farm less rope than another. What I can't independently confirm is which package version the live competition actually runs; this is read from `kaggle-environments 1.32.7`. If a season ever behaves as though it started with a different number, that's the first thing to check.

## 2. The Land and the Shed

Your farm is 100 acres, a 10×10 grid split into four 25-acre quarters. On day 1 you own only the top-left (NW). The other three have locked gates — your men can walk through them as a shortcut, but can't plant anything until you buy the deed: **\$1,000 for the NE**, **\$2,000 for the SW**, **\$4,000 for the SE**, in that exact order.

Right in the middle sits your shed. It holds exactly **100 items**. Seeds don't count — they stay in your pocket. But here's the secret: **your men's hands are uncapped**. A man can carry a whole field's harvest. So don't

shuttle to the shed out of habit; walking is the single biggest expense on this farm. Hold the crop in hand, and only drop it when a whole field ripens at once, or when you're about to sell.

Why? Because at midnight, whatever the men are carrying gets dumped into the shed. If that pushes the shed past 100 items, the overflow is thrown in the trash.

*A note on that: when I ran our own farm and counted the hours, roughly half of every man's day went to walking and only about a fifth to actual work. That's a measurement of how we were running things, not a law of the land — a better-routed farm would do better. But it tells you where the money leaks.*

### 3. The Crew, and the Midnight Vanishing Act

You have one permanent farmer. To get real work done you hire day-laborers, and they are nearly free — the price follows a Fibonacci ladder that resets every morning:

| Hand of the day | 1   | 2   | 3   | 4   | 5    | 6    | 7    | 8    | 9    | 10    |
|-----------------|-----|-----|-----|-----|------|------|------|------|------|-------|
| Cost            | \$1 | \$1 | \$2 | \$3 | \$5  | \$8  | \$13 | \$21 | \$34 | \$55  |
| Running total   | \$1 | \$2 | \$4 | \$7 | \$12 | \$20 | \$33 | \$54 | \$88 | \$143 |

Hiring ten hands costs **\$143 for the whole day** — that is the ten hired hands only; your permanent farmer is free and always there, so the farm runs on eleven pairs of hands in total. The catch: every night at midnight they all go home, the hiring ledger resets to zero, and you must call the hiring hall again each morning. **Not hiring is the most common way bosses leave money on the table.**

## 4. The Crops — Watering is the Paycheck

Water on planting day, every time. A seed starts with one “missed water” mark the second it hits the dirt. If you don't water it that same day, it hits two marks at midnight and turns into a weed. After that you may skip at most one day between waterings.

There are two kinds of crops, and they want opposite things from you.

### One-time crops — watering builds the harvest

| Crop   | Seed | Full harvest | Watering window           | No fertilizer | Fertilized max | Barely alive |
|--------|------|--------------|---------------------------|---------------|----------------|--------------|
| Wheat  | \$10 | day 4        | days 2–4                  | 4             | 6              | 1            |
| Carrot | \$20 | day 3        | days 2–3                  | 3             | 4              | 1            |
| Melon  | \$80 | day 10       | days 6–10 (11–12 nothing) | 6             | 6              | 1            |

The fifth column is what the office actually gets — **it never uses fertilizer**, so treat that as the real ceiling for this farm, not the game's. Fertilizer doubles the watering bonus (2 units instead of 1), which is exactly why the true engine maximum in the sixth column is higher for wheat and carrot — melon's window is already long enough to reach its cap without it. If a future boss adds fertilizer, wheat and carrot both have real headroom sitting unused.

#### A word on the melon — I had this wrong myself

She is *ready* at day 10, not day 12. Watering her on days 11 and 12 adds nothing; she is already full. So **day 19 is the last day you can plant a melon and still cash it**, not day 17 as I first told you. That is two extra days of our best crop I nearly threw away.

One honest caveat: day 19 is the exact edge, not a safe margin. A melon planted then first becomes harvestable on the very last day of the season — and if the harvest itself lands late in that day's hours, there may be little or no time left to drop and sell her. This is the same last-hour-harvest limit measured elsewhere in this book (see “Where the Farm Stands Today”): most seasons clear the shed completely, but the worst measured case left \$1,749 unsold for exactly this reason. Planting a day earlier costs nothing and buys a full day of slack; the office does not currently do this.

The same thinking cuts the other way for wheat and carrots. The figures above are for a *full* harvest. If you'll take a smaller one, both are pickable from day 2 — so in a pinch you can plant either through day 27 and still turn a bushel into cash.

### Repeaters — the four-fruited death clock

- **Tomato (\$50 seed)** — first fruits on day 8, then every 1 day.
- **Strawberry (\$100 seed)** — first fruits on day 10, then every 2 days.

For these, water is *just survival*. They fruit on their own calendar — 4 times total, whether you pick them or not. After the fourth fruiting they dry up and turn to weeds. Picking costs nothing and tires nothing, so pick whenever a man is passing by. Don't hold them on the vine to “save” the plant; the plant doesn't wait for you.

**Weeds.** Bare ground doesn't stay bare. Any empty square can sprout a weed overnight — about one square in two hundred each night — and a plant you let die of thirst becomes one too. A weed is dead ground: nothing grows there until a man walks over and digs it out. Digging costs an hour and yields nothing, so it sits low on the crew's list — but leave weeds long enough and you are farming a smaller farm than you paid for.

**Fertilizer.** When men scoop manure from animals it becomes fertilizer. Applied to a crop it doubles the watering bonus — 2 units per watering instead of 1 — for exactly 3 days. That's why we scoop it daily; left in

the yard overnight it's wasted.

## 5. The Animals — the Long Game

**Read this chapter, but know what it is**

Everything below is true of the farm, and I have written it out so the next boss can put it to work. But **today's brain does not keep animals** — nobody has written the build-buy-carry-place-feed round yet. It is the biggest thing left undone, and this chapter is here to make it easy to pick up, not to describe what happens now.

Names matter: **geese live in a COOP, cows and sheep in a PASTURE**. There is no barn and no chicken.

It is a four-step chore with a hidden fifth: a man builds the home on empty dirt (free); you phone the market and buy the animal, which **arrives in the shed**; a man walks to the shed, picks it up, carries it across the farm and places it in its home; then every single day a man must feed it one bushel of wheat **from his own hands** — he cannot feed it from the shed, so he must walk to the shed, pick up wheat, carry it over, then feed. Miss two days and the animal escapes forever. While he's there, have him CARE for it; that banks a bonus paid on her next laying day.

| Animal | Cost  | Home    | Produces | Starts | Then every |
|--------|-------|---------|----------|--------|------------|
| Goose  | \$300 | COOP    | Egg      | day 4  | 1 day      |
| Cow    | \$400 | PASTURE | Milk     | day 8  | 2 days     |
| Sheep  | \$500 | PASTURE | Wool     | day 6  | 3 days     |

Unlike a crop, an animal **never stops producing** as long as she is fed. In one good run a goose laid **52 eggs** in a season — treat that as a measurement of what's possible, not a law. And don't expect a gold mine: once you count the walking and the wheat, an animal clears maybe **15% more per hour of a man's time** than plain wheat would have. A steady edge that lasts all season, but earned the hard way.

## 6. The Market — the Ten Phone Lines

To sell you don't send a man to town. You phone the commodity broker from the office — but he only takes **10 calls an hour**. Each call can be as big as you like; after the tenth he stops listening. And you can only sell what is **already in the shed**, never what a man is still holding. You can also buy seeds at fixed prices; they go straight to your pocket and never touch the market.

The broker sells in the other direction too, but only two things: **wheat and fertilizer**. Everything else he'll buy from you and never sell back. Wheat is the useful one — it means you can keep a cow fed without giving up a single square to grow her feed. Buy the wheat, it lands in the shed, a man picks it up on his way past, and the pasture never costs you a planting. Two cautions: bought goods land in the shed like everything else, so they count against the 100, and he won't sell you a thing if the shed is already full.

The market is shared with your neighbour, and every product has its own crash curve. Two numbers matter for each: where the price **halves**, and where it hits the **\$1 floor**. They are not the same number, and mixing them up will cost you.

| Product | Normal price | Curve | Halves after | Hits \$1 floor after |
|---------|--------------|-------|--------------|----------------------|
| Wheat   | \$25         | log   | never        | never                |

| Product    | Normal price | Curve  | Halves after | Hits \$1 floor after |
|------------|--------------|--------|--------------|----------------------|
| Egg        | \$50         | log    | never        | never                |
| Carrot     | \$35         | sqrt   | 230 units    | <b>842 units</b>     |
| Tomato     | \$60         | sqrt   | 135 units    | <b>529 units</b>     |
| Melon      | \$250        | sq     | 112 units    | <b>158 units</b>     |
| Milk       | \$160        | linear | 38 units     | <b>76 units</b>      |
| Strawberry | \$120        | linear | 31 units     | <b>62 units</b>      |
| Wool       | \$200        | sq     | 42 units     | <b>59 units</b>      |

Wheat and eggs are your bottomless money pipes — sell two thousand bushels of wheat and the last one still fetches about \$19. Melon is far sturdier than she looks in that bottom group: she absorbs nearly three times what wool does before she's worthless. The price never falls below \$1, but at \$1 you're giving it away.

## 7. The Town and the Neighbour

Here's the upside rookies miss. The town is always eating, from two mouths:

- **The town centre** takes one of every product every 24 hours — once a day, flat, all season.
- **The shops** are the real appetite. A new one opens every 3 days, up to 8 in total, and each open shop eats every 4 hours — **6 units a day of every product it wants**. Two separate things can double that, and they stack: a shop that wants only one product (Pet Cafe and Yarn Store) eats **12 a day** of it instead of 6; and because shops are drawn with replacement, the same shop can open more than once, with **each copy eating its own full share**. Two Pet Cafes means 24 carrots a day, and you might end up with three bakeries and no yarn store.

By late season the shops eat many times what the town centre does. That draining pushes prices *up*. A tomato that normally fetches \$60 can fetch **\$900** late in the season if nobody is supplying it.

### Taught here, not yet done

The next two ideas — planting for scarcity you predict, and watching what the neighbour is planting — are both true of the farm, and both were built, measured, and **switched off because they made us poorer**. The reasons sit beside them in the rule book. Learn them; just do not assume the brain is doing them today.

### The trap in that

By the time you see the \$900 price it is too late to plant one — a tomato takes 8 days to fruit. **You must plant for the scarcity you predict, not the scarcity you see.**

And you aren't farming alone. You're racing one neighbour, scored head to head. You can see his money, his tiles, his men's positions and his land — but not his shed or his seeds. When two good farms sell into the same market both take a real haircut; in our own trials two evenly matched farms each ended up roughly a quarter poorer than either would have been alone. So use your eyes: see what he's planting, don't be the second man flooding melons, and race him for the premiums his men can't reach.

## 8. The Golden Rule of the Endgame

Because the bank only counts cash, the real rule for planting is simple: **plant it only if it'll be ready before the auditor arrives.**

| Crop       | Needs                  | Last day for a FULL harvest | Last day for ANY cash |
|------------|------------------------|-----------------------------|-----------------------|
| Carrot     | 3 days                 | day 26                      | day 27                |
| Wheat      | 4 days                 | day 25                      | day 27                |
| Tomato     | 8 days to first fruit  | day 21                      | day 21                |
| Strawberry | 10 days to first fruit | day 19                      | day 19                |
| Melon      | 10 days                | day 19                      | day 19                |

Two different questions, so two answers. The third column is when to stop if you want the crop's full yield — that is what the office actually uses. The fourth is the true last gasp: wheat or carrot can be pulled from day 2 for a single unit, so with idle hands and bare ground on day 27 there is still a bushel in it. The office does not take that gamble today; it plants for the full harvest or not at all.

There's no magic date on the calendar. You ask the question crop by crop, every morning. And once the answer is “no” for everything, you stop planting, stop buying land, stop buying animals, and switch to pure liquidation: harvest everything, have the men carry it to the shed and drop it, and phone the broker until it's all cash before hour 720.

## 9. The Office Ledger — part built, part intended

Some of this already runs. The office does compare each evening's takings against that morning's guess and adjusts how far it trusts itself. It does count the plants that went unwatered and shrink or widen the acreage accordingly. And the checking tools already test the brain against an empty farm, a bare register and a full shed before anything is handed over.

Still only intention: a proper ledger that sorts our mistakes by kind — was it the market, the growing, or the warehouse that let us down — rather than one running score; and a hard limit on how many times we re-think a decision inside a single hour.

I'll be honest that the last one is belt-and-braces: our office answers the phone in about two thousandths of a second against a full second allowed. But the penalty for one slow answer is the whole season, and I'd rather wear the belt.

# The Boss's Day, on One Page

Pin this by the map. Every hour, for every man, walk down the ladder — the first “yes” wins, and a dying plant always outranks a clever plan.

## EVERY MORNING (hour 0)

*before anyone takes a step*

**Re-hire the whole crew**

*They all went home at midnight.  
Ten hands cost about \$143 for the day.*

Phone the broker: sell what is in the shed

*Only 10 calls an hour. Only shed goods.  
Never what a man is still carrying.*

## EVERY HOUR — one job per man, in this order

*first YES wins*

Is anything dying today?

YES

**Water the plant. (Feed the animal.)**

*Two missed days and it is gone. Feeding: not built yet.*

NO

Is anything ripe and full?

YES

**Pick it.**

*One-time crops: wait for the full window first.*

NO

Will the shed overflow tonight?

YES

**Carry the load in and drop it.**

*Overflow past 100 items is binned at midnight.*

NO

Is there empty ground worth planting?

YES

**Plant it — and water it today.**

*An unwatered seed dies the very same night.*

NO

Walk one step toward the next job

*Walking is the biggest cost on the farm.*

Is he standing ON the square?

NO → step toward it

*Boots in the dirt. No chore works from the edge.*

## AT MIDNIGHT

*nothing you can do about it then*

Hands go home • carried goods dumped into the shed, overflow binned • plants that missed two waterings turn to weeds

## AND EVERY MORNING OF THE LAST WEEK

*asked crop by crop*

Can it be CASH by hour 720?

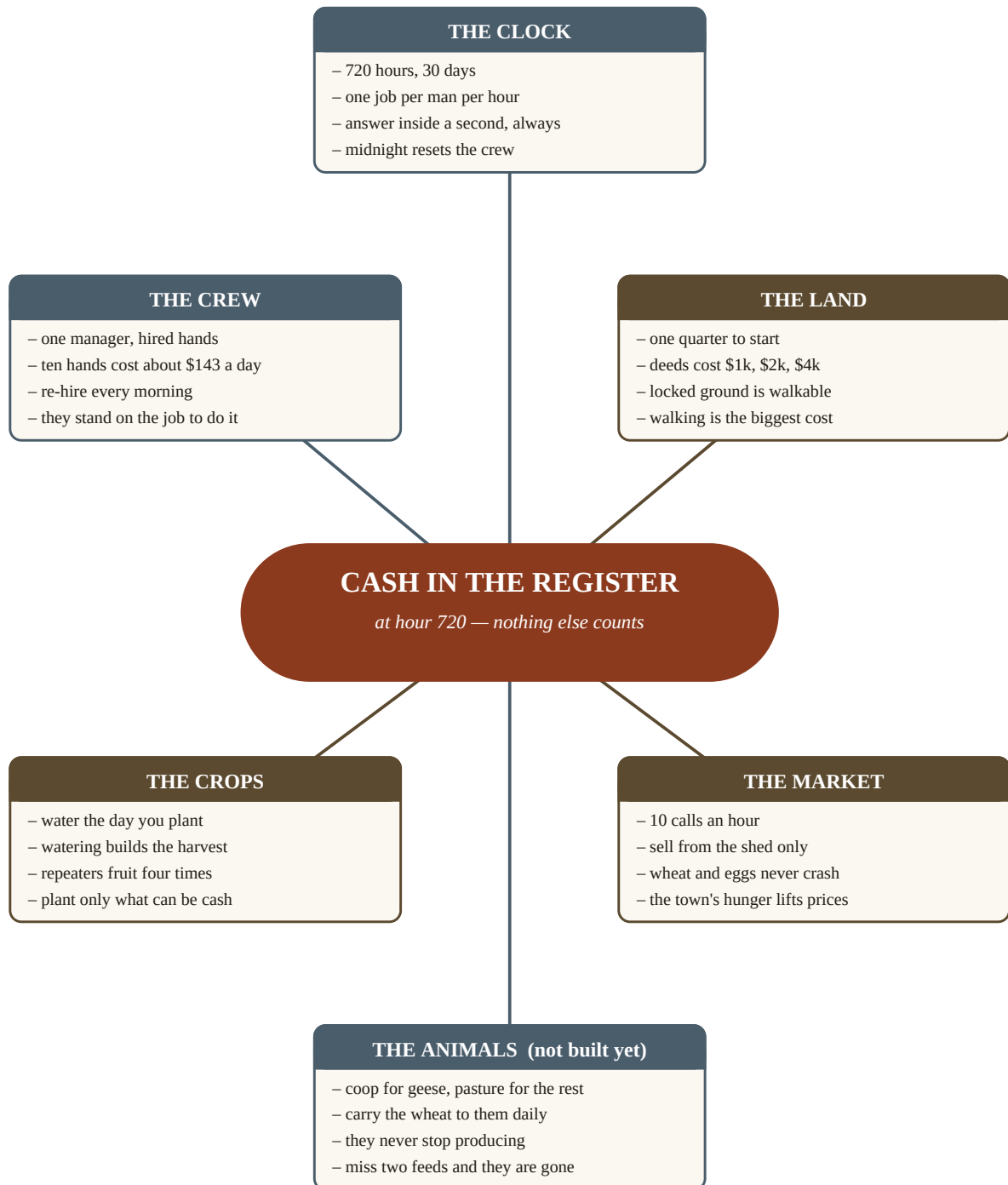
NO

**Stop planting and buying. Harvest, drop, and sell everything.**

*FULL-HARVEST policy: carrot 26 • wheat 25 • melon 19  
(a policy, not a hard cutoff — wheat/carrot last-gasp is day 27, unused)*

# The Farm at a Glance

One page, six things to keep in your head. Everything on it points at the middle.





# The Standing Orders

The same instructions the office follows every hour, written out the way you'd say them aloud. No jargon — if a line here doesn't make sense to you, it doesn't belong in the farm's brain either.

One line below is marked a hunch. When the season looks like ending poorly, the office works more land and sells cheaper, on the theory that a poor season is no time to be choosy. It is the only instruction in this book I never measured — in a normal season it has never once fired, so I have no evidence either way. Treat it as an opinion I left behind, not a finding.

## EVERY MORNING, BEFORE ANYONE MOVES

Call the hiring hall until the crew is back up to strength.  
Phone the broker and sell whatever sat in the shed overnight.

## EVERY HOUR

Work out what the farm will be worth on the last day.  
If that number looks poor, widen the acreage and accept  
lower prices to keep the cash moving. (a hunch —  
never measured; see the note above)  
If the register is nearly empty, sell now and buy nothing.

## THEN, FOR EACH MAN IN TURN — first yes wins

|                                   |                                    |
|-----------------------------------|------------------------------------|
| Is anything of mine dying today?  | → go water it (feeding: not built) |
| Is anything ripe and fully grown? | → go pick it                       |
| Will the shed overflow tonight?   | → carry the load in and drop it    |
| Is there ground worth planting?   | → plant it, and water it today     |
| Otherwise                         | → step toward tomorrow's work      |

If he is already standing on the square, he does the job.  
If he is not, he takes one step toward it and we ask again next hour.

## BEFORE PLANTING ANYTHING, ASK

Which crop is worth most per square per day at today's prices,  
counting what I already have coming to market?  
Can it be harvested and sold before the auditor arrives?  
If no crop passes both tests, plant nothing.

## EVERY EVENING

Compare what the farm actually earned today against what I  
predicted this morning, and trust my next guess accordingly.  
Count how many plants went unwatered.  
If any did, work less land tomorrow.  
If the men stood idle instead, work more.

## IN THE LAST WEEK — same crop-by-crop test, nothing new

Stop buying land. Stop buying animals.  
Keep planting whatever can still be cash — ask crop by crop,  
every morning. Only when none pass do we stop planting.  
Send every man to the shed to drop what he is carrying.  
Sell everything, at any price, before the clock runs out.

## The Rule Book on the Shelf

Every figure the brain uses about the *world* lives in one file, `rules_validated.json`, and the brain looks each one up when it needs it. That file is the rule book on the office shelf. It holds three kinds of page, and a fourth kind of number never appears in it at all — knowing which is which is the whole trick.

| Kind of page       | What's on it                                                                                                                                                                                                           | May you change it?                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The laws</b>    | Seed costs, growing times, animal prices, the land deeds, the shed's 100-item limit, the shape of every price curve. These are facts about the world.                                                                  | <b>No.</b> Change one and the office starts believing something false. They are copied from the official rulebook and checked against it. |
| <b>The chores</b>  | Every recurring job — watering, feeding, caring, picking, planting, digging, dropping — with how often it comes round and how loudly it shouts.                                                                        | <b>Carefully.</b> Reordering these changes what the men do first when they can't do everything.                                           |
| <b>The derived</b> | Nothing is written here at all — these are worked out fresh every hour: what each crop is worth today, what the farm will be worth at the end, how much land the crew can tend, how far to trust this morning's guess. | <b>Nothing to change.</b> They follow from the other three. If a derived number looks wrong, the fault is in a law or a dial.             |
| <b>The dials</b>   | How big a crew to hire, how cheap a price you'll accept, how much land to work, when to start selling off. Judgement calls, not facts.                                                                                 | <b>Yes — this is what you're meant to turn.</b> Nothing here comes from the rulebook; it's all opinion, and it can be improved.           |

### Page by page, and what each one governs

| Section in the rule book | Kind  | What it decides                                                                                               | Told to you in                 |
|--------------------------|-------|---------------------------------------------------------------------------------------------------------------|--------------------------------|
| constants                | law   | Season length, shed size, 10 calls an hour, land prices, the hiring ladder, two missed waterings kill a plant | §1, §2, §3                     |
| crop_params              | law   | Seed cost, when each crop first bears, when it stops improving, how many it gives                             | §4                             |
| market_params            | law   | Every price curve — which crops crash and which absorb                                                        | §6                             |
| town_model               | law   | Which shop eats what, how often, how many can open                                                            | §7                             |
| animal_params            | law   | Cost, home, first yield and rhythm for each animal ( <i>reserved — the brain doesn't keep animals yet</i> )   | §5                             |
| daily_routines           | chore | The nine recurring jobs and how loudly each shouts. A plant about to die shouts loudest of all.               | §4, §5, the daily routine page |
| policy                   | dial  | Crew size, how low a price to accept, how much land per man, when the last week begins                        | §3, §6, §8                     |
| endgame / forecast       | dial  | When to stop planting; how much to trust this morning's guess                                                 | §8                             |
| ops                      | law   | The exact words the walkie-talkie understands. Used by the checking tool, not by the brain.                   | §1                             |

### The pages I tore out

When I took over, the rule book had eleven more sections in it that *nothing in the brain ever read* — an old set of chores, an old set of targets, a list of forty-nine action rules. Harmless clutter, you'd think. It wasn't. Five of those dead entries **contradicted the live ones**, and when an inspector came through he read the dead numbers and wrote up the farm for settings we don't actually use:

| The dead page said                      | The farm actually uses                  |
|-----------------------------------------|-----------------------------------------|
| strategy.min_terminal_target = 3,000    | constants.MIN_TERMINAL_TARGET = 35,000  |
| strategy.correction_threshold = 2,600   | constants.CORRECTION_THRESHOLD = 45,000 |
| endgame.enter_liquidation_at = turn 640 | policy.endgame_start_turn = turn 670    |
| priorities.WATER = 1,100                | daily_routines.WATER.priority = 5,000   |
| priorities.FEED = 1,100                 | daily_routines.FEED.priority = 8,800    |

They're now shut in a drawer at the back of the book marked `_inherited_unused`, with a note saying nothing reads them. I kept them rather than burning them, but if you never miss them, burn them.

**One rule book, two copies — and why that matters**

The brain carries a copy of the rule book sewn into its own lining, so it still works if the shelf is bare. It prefers the one on the shelf if it finds it. That means **editing the shelf copy alone is not enough** — you must rebuild, or the brain may go on reading the old copy in its lining. The build tool refuses to hand you a package where the two disagree.

# How the Office Reads the Rule Book

Written out the way you'd say it aloud. This is the whole lifecycle: check it, find it, use it, and prove afterwards that it was the one you meant.

## BEFORE THE SEASON — checking the book against the real world

Open the official rulebook that came with the game.  
For every number in our book, find the same number in theirs.  
If ours disagrees, ours is wrong. Correct it and say so out loud.  
If theirs has something ours never mentioned, copy it across.  
Refuse to go any further while a single number still disagrees.

## WHEN THE OFFICE OPENS — finding the book

Look on the desk we were handed. Is the rule book there?  
If not, look in the room we happen to be standing in.  
If still not, read the copy sewn into our own lining.  
Before trusting whichever we found, flick through it:  
are all the pages we depend on actually present?  
If pages are missing, put it down and use the lining copy.  
Write down which copy we ended up using, so anyone can check later.

## EVERY HOUR — using it

Never remember a number. Look it up, every single time.  
How urgent is watering? → ask the chores pages  
What will this crop fetch? → ask the price-curve pages  
How many hands should we hire? → ask the dials  
Has the last week begun? → ask the dials

## WHEN SOMEONE WANTS TO CHANGE SOMETHING

Turning a dial is welcome. Rewriting a law is not.  
After any change: check the book against the official rulebook again,  
then rebuild, so the copy in the lining matches the copy on the shelf.  
Refuse to hand over a package where the two copies disagree.

## AND THE RULE THAT COVERS EVERYTHING ELSE

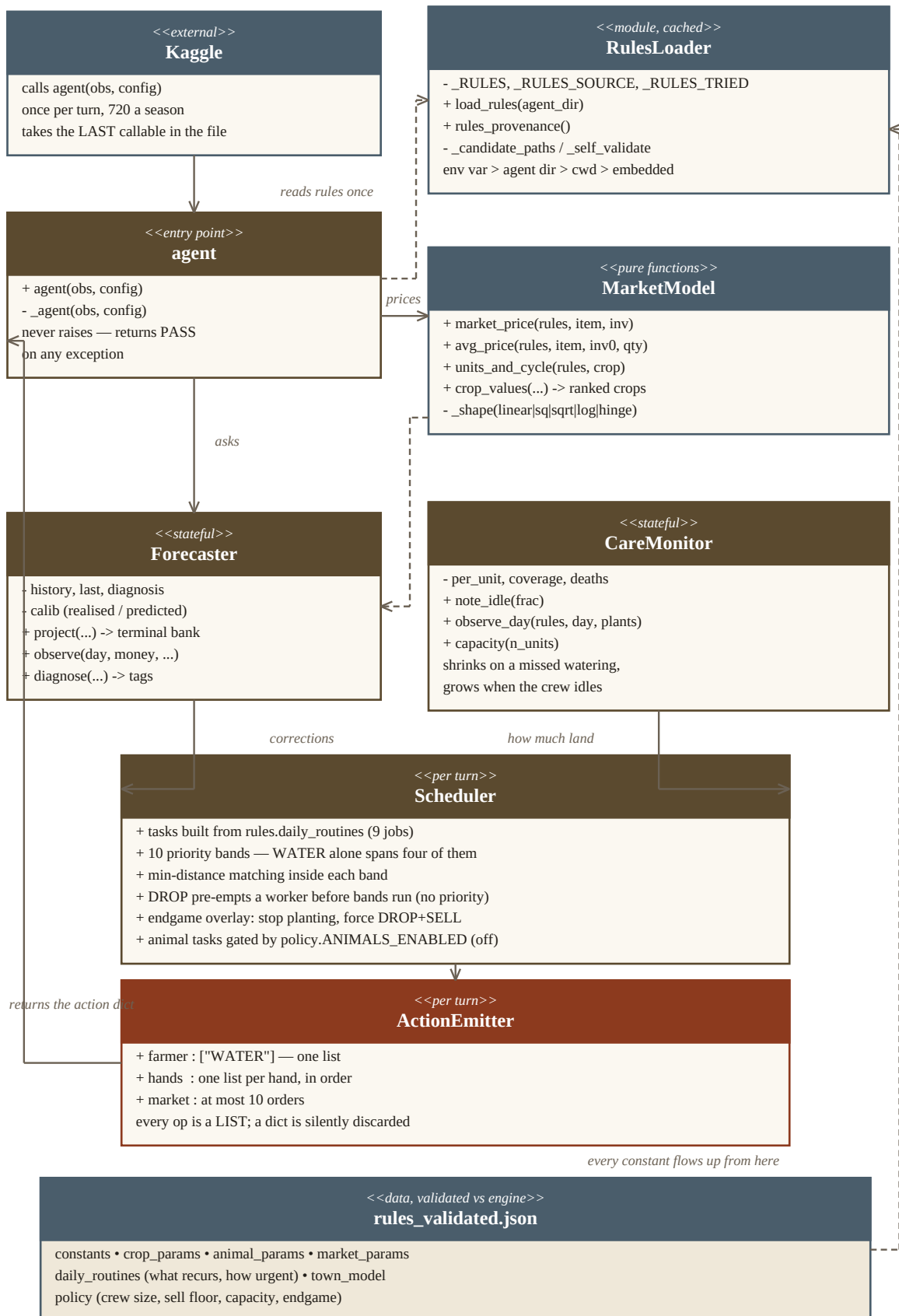
If our book and the official rulebook ever disagree,  
the official rulebook wins. Always. Without argument.

## How the Brain Is Built

For whoever writes the code rather than runs the office — the same farm, drawn as the parts of the program. Solid arrows are calls; dashed arrows are reads. The principle is that **no game-domain constant is duplicated in the strategy code** — seed costs, growing times, price curves and land prices exist only in the rules file. The code does still carry 36 numeric fall-backs — `.get(key, default)` call-sites, not the count of keys they cover. A single mandatory key can be read from more than one place; several dial names can share one line. They are not all the same kind of thing, and the difference decides whether a bad rule book fails loudly or quietly. This table is generated from the code each time this document is rebuilt, so it cannot go stale the way a hand-typed count can.

| Kind of fall-back            | Fallback sites | What happens if the rule book lacks that key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A mandatory game constant    | 5              | <b>Cannot happen — the fallback value itself is unreachable.</b> 5 of the 36 call-sites read a mandatory key with a numeric fallback, but none of them ever fire, because 12 mandatory keys in total — season length, shed size, order cap, land prices, the hiring ladder, the price floor, the two-missed-waterings rule and the rest — are checked before any of that code runs. If one is missing, only <i>that external file</i> is set aside; the office does not stop for the day. It reads the copy sewn into its own lining instead and carries on the hour exactly as before. Nothing about the run itself fails. This count is call-sites, not keys; the mandatory list itself has 12 entries. |
| A policy dial                | 22             | The default applies — intended: an unset dial should behave sensibly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| A field of the hourly report | 9              | Not a rule at all — defensive reads of what the game hands us (day, hour, yield so far).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Genuinely unguarded          | 0              | <b>None remain.</b> There were two once: <code>LIQUIDITY_RESERVE</code> was a tuning constant nobody had checked — it is mandatory now, counted in the first row — and forecast smoothing turned out to already be a policy dial, counted in the second. The gap this row used to name is closed.                                                                                                                                                                                                                                                                                                                                                                                                         |

The rule that matters, stated precisely: **an external rule book missing one of the 12 mandatory keys is set aside, never quietly patched with a made-up number — but the office itself does not stop.** It falls back to its own sewn-in copy and keeps working the same hour. “Rejected” describes the file, not the farm; the run is unaffected as long as the sewn-in copy is sound, which the build tool checks before it ever ships one. One file missing a dial is simply accepted, defaults and all. What counts as mandatory is a list beside the loader in the code, so the distinction is checkable rather than a promise.



## Every Instruction, and Where It Lives

Everything I told you in this book is written down somewhere in the farm's brain. Here is the map from one to the other — what the office does, which file makes it happen, and which tool proves it still works.

### Things the farm does today

| What I told you                                  | Where it lives                                                               | What proves it                                          |
|--------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------|
| Re-hire the whole crew every morning             | main.py places HIRE orders at hour 0; how many is <i>policy.target_hands</i> | compliance_audit.py counts the orders                   |
| The man must stand on the thing he works on      | main.py walks him there a step at a time, then acts                          | compliance_audit.py flags any job done off-square       |
| Water on planting day, never miss twice          | <i>daily_routines.WATER</i> — a plant about to die outranks everything       | revalidate.py reports watering coverage                 |
| Work only as much land as the crew can tend      | CareMonitor: shrinks after a missed watering, grows when men idle            | revalidate.py counts plants lost to thirst              |
| Watering builds the harvest, so don't pick early | main.py holds a one-time crop until its window closes                        | the crop table is read from the engine                  |
| Pick repeaters whenever a man is passing         | harvested as soon as the tile is full                                        | compliance_audit.py finds no wasted picks               |
| Don't let the shed overflow at midnight          | men are sent in to drop early once the load nears the cap                    | revalidate.py; median 0, worst 7 of 100 over 20 seasons |
| Sell from the shed, ten calls an hour            | the market list, capped at 10                                                | compliance_audit.py; 0 violations in 62,545 turns       |
| Don't flood a crop that crashes                  | crops are priced against live market stock plus our own crop still growing   | the price curves are read from the engine               |
| Plant only what can be cash in time              | every crop is tested against the days remaining, every morning               | the cutoffs come from the engine                        |
| Sell everything before the clock runs out        | <i>policy.endgame_start_turn</i> , then forced drop-and-sell                 | usually clears the shed; see the honest exception below |
| Answer inside the second, never crash            | the whole turn is wrapped so nothing can throw                               | revalidate.py measured 3 ms against a 1,000 ms limit    |

#### The honest exception to “sell everything”

Almost always the shed clears completely. But if a harvest first becomes sellable on the **very last playable hour** — step 719, with no step 720 to follow — there is no turn left in which to phone the broker. Across 20 measured seasons that leaves a **median of 0 items, but as many as 7 in the worst case, worth up to \$1,749**. This is not a bug to fix; it is a hard edge of the clock itself. “Sell everything” is the right instruction and it is what the office does — it just cannot always be finished with zero turns left to do it in.

### Things I told you about that the farm does not do yet

| What I told you                   | Why it isn't in the brain                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Keep a goose and she keeps paying | The build-buy-carry-place-feed chain is written into the rules file as a routine, but no code runs it yet. This is the biggest thing left undone. |

| What I told you                        | Why it isn't in the brain                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scoop the manure for fertilizer        | Waits on the animals. Without them there is no manure, and buying it outright hasn't paid for itself in trials.                                                                                         |
| Watch what the town will be hungry for | I built it and measured it, and it made us <i>poorer</i> — the town's daily nibble is too small to matter over one crop's life. Left in the file, switched off, with the measurement written beside it. |
| Watch what the neighbour is planting   | Same story: built, measured, made us poorer. The market price already tells us what he has sold, so counting his standing crops as well double-counts him.                                              |

I've left the last two switched off rather than deleted, with the numbers beside them, so the next boss doesn't spend a week rediscovering that they don't work.

## Where the Farm Stands Today

A handover with no numbers in it is a handover you can't audit. Here is exactly how well the farm runs, measured rather than asserted, so the next boss can tell whether a change helped.

One honest note before the numbers: the release bar (MIN\_TERMINAL\_TARGET) is **35,000**, calibrated from these very measurements, not picked in advance and left soft. It sits below every measured mean and close to the worst measured season *against idle and starter neighbours* (\$36,526 and \$38,311) — that's the comparison it was calibrated against, so that's the honest reading. Self-play's worst case (\$17,747) sits well under the bar and always will: two identical farms splitting one market is a different measurement, not a target this bar was ever meant to clear. Setting the bar at the mean would reject half of normal play as a failure, which is the point of setting it near the worst case instead. The crop logic below is doing real work compared with a single-crop bot, but the final score still depends on the calibration holding up and on full-season execution, not on this table alone.

### What a season is worth

| Neighbour across the road     | Seasons | Mean     | Median   | Worst    | Spread (sd) | Won      |
|-------------------------------|---------|----------|----------|----------|-------------|----------|
| An idle farm (does nothing)   | 25      | \$50,458 | \$49,023 | \$36,526 | \$8,711     | 25 of 25 |
| The starter farm (one square) | 25      | \$54,838 | \$57,040 | \$38,311 | \$10,153    | 25 of 25 |
| A copy of ourselves           | 20      | \$37,234 | \$37,852 | \$17,747 | \$7,436     | 7 of 20  |

Read the last row carefully. Two identical farms selling into the same market both end up about a quarter poorer than either would alone — that is the honest number for a real contest, not the \$50,000 you get against an idle neighbour. Winning 7 of 20 against a copy of yourself is not literally a coin toss — but it's well within the range a true 50/50 matchup would produce by chance (flip a fair coin 20 times and it lands 7-or-fewer or 13-or-more heads about one time in four — that two-sided spread, not the one-sided “7 or fewer” alone, is the right comparison for asking whether a result looks unusual), so it's consistent with a roughly even matchup, which is what two identical farms should be.

### What else was measured, and under what conditions

| Measurement                       | Value                                   | How it was measured                                     |
|-----------------------------------|-----------------------------------------|---------------------------------------------------------|
| Time to decide one hour           | mean 1.7 ms, 99th 3.8 ms, worst 50.9 ms | 9 seasons, 62,545 decisions; the limit is 1,000 ms      |
| Illegal orders                    | 0                                       | 9 seasons vs idle, starter and random neighbours        |
| Wasted orders (legal but useless) | 0                                       | same 9 seasons, every order checked                     |
| Watering kept up                  | 93.2% of plant-days                     | 1 season, seed 42, counted at the last hour of each day |



| Measurement                  | Value                             | How it was measured                                                |
|------------------------------|-----------------------------------|--------------------------------------------------------------------|
| Plants lost to thirst        | 10 per season                     | same season; target is zero, we are not there                      |
| Goods left unsold at the end | median 0, worst 7 (up to \$1,749) | 20 seasons vs an idle neighbour; see the last-hour exception above |
| Time spent walking           | about half of all worker-hours    | 1 season, seed 42; work 22%, idle 25%                              |

All figures from kaggle-environments 1.32.7 at default settings, 720 turns a season. Anything measured on a single season is marked as such — treat those as indications, not constants.

### What counts as an improvement — the number that stops arguments

The spread between seasons is about **\$8,700**. Across 25 seasons that leaves roughly **\$1,740** of uncertainty on the average. So: run **at least 25 seasons on the same seeds, before and after**, and accept the change only if the average improves by more than **\$3,500** — twice the uncertainty. That number is not a law: it is  $2 \times (\text{spread} \div \sqrt{\text{seasons}})$ , so recompute it whenever the spread or the season count changes. Better still, run the *same* seeds before and after and look at the average difference, which cancels part of the luck.

### Why that is the right shape of rule

The season is **deterministic** — the same farm on the same seed finishes with the same cash to the dollar. Every difference between seasons comes from the seeds, not from luck on the day, so a comparison must be run **on the same seeds before and after** and judged on the *difference per seed*, never on two separate averages.

You would expect matched seeds to cancel the noise entirely. They don't, and the reason matters: change one dial and the farm plants differently, which moves the market, which changes every later decision — the two seasons drift apart. Measured on a real change, ten hands against nine over fifteen matched seeds:

| What you measure                      | Spread  | Uncertainty on the answer |
|---------------------------------------|---------|---------------------------|
| Two separate averages (unpaired)      | \$8,305 | \$2,144                   |
| The difference, seed by seed (paired) | \$5,933 | <b>\$1,532</b>            |

Pairing buys about **1.4 times** the sensitivity — useful, not magic. At 25 matched seasons the bar lands near **\$2,400**; the \$3,500 above comes from the unpaired figure and is deliberately the stricter of the two. Worked example: ten hands against nine came out at **-\$656, uncertainty \$1,532** — not evidence that nine is worse, but evidence that fifteen seasons cannot tell them apart. Twice on this farm I mistook exactly that for an improvement. A change that lifts the *worst* season counts for more than one that lifts the average.

## What Works, What Doesn't

One table, so nobody has to piece it together from the rest of the book. Read **Working** as *built and running*, not as *flawless* — watering is marked working and still loses ten plants a season.

| The job                 | Status                                  | Note                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planting crops          | <b>Working</b>                          | Crop chosen fresh each hour on today's prices                                                                                                                                                                                                                                                                                                                                                                     |
| Watering                | <b>Working</b>                          | 93.2% kept up; 10 plants a season still lost                                                                                                                                                                                                                                                                                                                                                                      |
| Harvesting              | <b>Working</b>                          | Waits for the full window on one-time crops                                                                                                                                                                                                                                                                                                                                                                       |
| Hiring the crew         | <b>Working</b>                          | Ten hands re-hired every morning                                                                                                                                                                                                                                                                                                                                                                                  |
| Selling                 | <b>Working</b>                          | Priced against live stock and our own crop coming                                                                                                                                                                                                                                                                                                                                                                 |
| Buying land             | <b>Working</b>                          | Only once the farm runs out of ground to work                                                                                                                                                                                                                                                                                                                                                                     |
| Endgame liquidation     | <b>Working</b>                          | Usually clears the shed; worst case 7 items left (last-hour harvest, no turn left to sell)                                                                                                                                                                                                                                                                                                                        |
| Adaptive acreage        | <b>Working</b>                          | Shrinks after a missed watering, grows when men idle                                                                                                                                                                                                                                                                                                                                                              |
| Digging weeds           | <b>Working</b>                          | Low priority; weeds do build up late in a season                                                                                                                                                                                                                                                                                                                                                                  |
| Animals                 | <b>Not built — switched off in code</b> | No build-buy-carry-place-feed round. The feed and care chores <i>are</i> in the rule book, and generating them from an animal tile is behind a hard <code>ANIMALS_ENABLED</code> switch checked in the scheduler itself — off by default, tested both off and on. The release gate separately confirms zero animal ops in a normal season. Flip the switch, deliberately, the day the workflow is actually built. |
| Fertilizer              | <b>Not built</b>                        | Waits on animals for a free supply                                                                                                                                                                                                                                                                                                                                                                                |
| Town-hunger forecasting | <b>Built, measured, off</b>             | 25 paired seasons: <b>−\$4,037</b> a season (SE \$1,369); better on only 7 of 25                                                                                                                                                                                                                                                                                                                                  |
| Watching the neighbour  | <b>Built, measured, off</b>             | 16 paired self-play seasons: <b>−\$5,534</b> (SE \$1,345); better on only 2 of 16                                                                                                                                                                                                                                                                                                                                 |
| Counterfactual planning | <b>Not built</b>                        | The office never clones the farm to test 'what if'                                                                                                                                                                                                                                                                                                                                                                |

### The two switched-off ideas — all the evidence there is

Each was first judged on one narrow test, which was not enough to justify leaving it off. Both have since been re-run as matched-seed comparisons against a second kind of neighbour:

| Idea                    | Neighbour    | Seasons | With     | Without  | Difference       | Verdict        |
|-------------------------|--------------|---------|----------|----------|------------------|----------------|
| Town-hunger forecasting | idle         | 25      | \$46,421 | \$50,458 | −\$4,037 ± 1,369 | <b>harmful</b> |
| Town-hunger forecasting | starter      | 18      | \$45,923 | \$54,822 | −\$8,900 ± 2,031 | <b>harmful</b> |
| Watching the neighbour  | starter      | 18      | \$54,088 | \$54,822 | −\$735 ± 1,351   | no effect      |
| Watching the neighbour  | a copy of us | 16      | \$32,269 | \$37,803 | −\$5,534 ± 1,345 | <b>harmful</b> |

Town forecasting is harmful against both neighbours tried, so leaving it off is well founded. Watching the neighbour is the weaker case: it clearly hurts when the other farm is as busy as ours and does nothing measurable when it isn't. Neither has been tried across a wide mix of opponents or over longer runs, so *harmful here* is not *harmful always* — if the ladder puts us against very different farms, both deserve a re-test before being written off for good.

## Known weaknesses, in the order I would attack them

**1. Animals are missing entirely.** A goose laid 52 eggs in a test season and eggs never crash in price. Per hour of a man's time the edge over wheat is only about 15%, so it is worth doing but is not a windfall.

**2. Half the day goes to walking — an architectural limit, not a tuning knob.** Work is 22% of worker-hours, idling 25%, walking the rest. The office sends whoever is nearest within a priority band; it has no notion of a round, a territory, or a man finishing one corner before starting another. Every other improvement on this list competes for the fifth of the day that is actually productive — fixing routing changes the size of that fifth. I would rank it alongside the animals, not below them.

**3. Ten plants a season still die of thirst.** The target is zero. The acreage control trades a little land for reliability but has not closed the gap.

**4. Swallowed mistakes — now counted and gated.** The brain is still wrapped so nothing can crash a season: on any error it does nothing that hour. But every catch is now recorded with the hour it happened, and the release gate **fails unless a full season swallows nothing at all**. A quietly broken farm can no longer pass for a healthy one. Current count over a full season: **zero**.

**5. The release checks don't name what they cover.** The 28 acceptance checks pass, but none of them prints which original checklist requirement it satisfies. “All checks passed” therefore proves the 28 things we thought to test — not that every required behaviour is covered, and nobody can tell from the output which requirements have no test at all. Until each check carries its requirement IDs, treat a green run as evidence, not proof.

**6. The market is starved, not flooded.** Late-season prices finish *above* normal almost everywhere — tomato at \$900 against a usual \$60. We are not over-supplying; we are failing to supply. That points at throughput, which points straight back at items 1 and 2.

## What I'm Handing Over

Everything in the filing cabinet, boss, and what each thing is for. Only the first item goes to the competition; the rest are the tools that keep it honest.

### The submission

| File                 | What it is                                                                                                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| submission.tar.gz    | <b>This is the one you send.</b> A tarball holding the two files below, with <code>main.py</code> at the root as Kaggle requires.                                            |
| main.py              | The farm's brain. Kaggle calls its agent ( ) function once per hour, 720 times a season. Holds no game numbers of its own — it reads them all from the rules file.           |
| rules_validated.json | Every game constant, every recurring chore and its urgency, and every dial worth turning (crew size, when to start selling, how much land to work). Edit this, not the code. |

### The tools that keep it honest

| File              | What it does                                                                                       | When to run it                                  |
|-------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------|
| validate_rules.py | Checks the rules file field by field against the real engine and fails loudly on any disagreement. | After editing rules, or on a new engine version |

| File                | What it does                                                                                                                                                | When to run it                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| build_submission.py | Rebuilds the tarball. Refuses to produce a broken one — it checks the embedded rules parse, match the source, and that Kaggle will pick the right function. | After <i>any</i> change            |
| revalidate.py       | 28 acceptance checks on the finished package: archive layout, offline running, rules discovery, latency, robustness, watering coverage.                     | Before submitting                  |
| compliance_audit.py | Plays full seasons and checks <i>every single order</i> against the official rulebook, separating illegal moves from legal-but-wasted ones.                 | Before submitting                  |
| mc.py               | Plays many seasons against different opponents and reports the spread, so a lucky run is never mistaken for an improvement.                                 | Before believing any change helped |

## The paperwork

| File                        | What it is                                                                                                                                                                                                                                                         |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kaggriculture_Handover.pdf  | This document. Plain-English guide to the whole business, with the daily routine on one page.                                                                                                                                                                      |
| Kaggriculture_Handover.md   | The same text in plain markdown, if you'd rather edit it than read it.                                                                                                                                                                                             |
| make_handover_pdf.py        | Rebuilds this document. Refreshes game-rule and code-fact figures automatically; performance figures do not update themselves — see the note at the very end of the book. Run it again after any engine change, but re-measure separately after a strategy change. |
| Kaggriculture_Explained.pdf | An earlier, more technical guide aimed at whoever writes the code rather than whoever runs the office.                                                                                                                                                             |

### The order to work in

Change the rules file, then `validate_rules.py`, then `build_submission.py`, then `revalidate.py`, then `compliance_audit.py`, then `mc.py` — and only submit if that last one shows a gain of more than \$3,500 across at least 25 matched seasons. Each step stops on failure, so the whole gate runs as one line:

```
python3 validate_rules.py rules_validated.json && python3 build_submission.py && python3 revalidate.py && python3 compliance_audit.py && python3 mc.py pass 25
```

And remember: the tarball carries its own copy of the rules, so rebuild it after every edit rather than swapping the file in by hand.

## In Short

You're the boss. You answer the walkie-talkie the moment it crackles — one slow answer voids the whole season. You hire cheap labour every morning, give short step-by-step orders, and make sure the man stands on the thing he's working on. You water seeds the day they're planted, and keep watering one-time crops through their window — that's the paycheck. You pick tomatoes whenever a man is passing; they die on their own clock. You dig the weeds out before they eat your acreage. And you turn every last crop into cold hard cash before the 30-day season ends.

The goose, the town's appetite and the neighbour's field are in this book because they are true and because the next boss should know them. The goose is honest fiction — nobody has written the code. The other two are real code that runs, sitting behind a switch turned off because the numbers said so. Not one of the three is *active in the current strategy*. Don't let a summary tell you otherwise.

I've got the keys, boss. Go enjoy your retirement.

This document has three kinds of number in it, and they don't all come from the same place or refresh the same way.

**Game rules** — crop costs, growing times, market curves, land prices, the hiring ladder, shed capacity and the rest — are read live from `kaggle_environments envs/kaggriculture` every time this document is built. They cannot drift from the engine; rebuild against a different engine version and they update themselves.

**Facts about the code** — the fallback classification on the architecture page, the count of mandatory keys, the number of acceptance checks — are computed from the actual source files at build time, the same way. They cannot drift from the code either.

**Measured performance** — the Monte Carlo means and medians, standard errors, self-play win rates, watering coverage, walking time, decision latency, the \$3,500 release threshold — is different in kind. These come from actually running the agent, and **rebuilding this document does not re-run it**: every one of those figures is a value transcribed from the last time someone explicitly ran `mc.py`, `revalidate.py` or `compliance_audit.py` and copied the result in by hand. If the strategy changes and nobody re-measures, these numbers will quietly go stale in exactly the way the game-rule and code numbers cannot. Treat anything under “Where the Farm Stands Today” as current only as of its last measurement, not as of the last rebuild.

## Revision 5 • Guardrails Closed and Open

The previous revision recorded three items as open rather than claim a fix that hadn't been made. Two of the three are now closed — not by asserting it here, but by an automated check that fails the release if either regresses. The third is still open, honestly.

**1. Endgame dates — unchanged, correctly a policy not a law.** The one-page guide still shows the office's current full-harvest cutoffs, not the absolute last-cash dates. Wheat and carrot keep a one-unit last-gasp option on day 27 that this strategy declines. No change needed; this was already stated correctly.

**2. Animal execution — now closed in the code itself, not only tested for.** The first attempt at this was a release check that confirmed no animal job ever appeared — a good test, but only a test: it would have caught the problem after the fact, not prevented it. The scheduler still generated FEED and CARE from any animal tile unconditionally; the check simply relied on no animal tile ever existing. That is exactly the gap the review meant by “until an explicit feature switch prevents those chores from entering the active queue.” It's fixed properly now: an `ANIMALS_ENABLED` switch in `rules_validated.json`, checked inside the scheduler *before* it looks at animal tiles at all, off by default. I proved it both ways: with an animal tile placed on the board and the switch off, the farmer standing on it returns `PASS`; with the same tile and the switch flipped on, he returns `FEED`. The release check still runs on top of that, as a second, independent layer — belt and braces, not just a belt.

**3. Swallowed exceptions — now closed by a counter, not a snapshot.** The wrapper still returns `PASS` on any exception so a season can never crash, but every catch is now recorded with the hour it happened, and `revalidate.py` fails unless a full season swallows **nothing at all**. This is no longer “we checked once by hand”; it is a standing gate that runs every time the farm is rebuilt. Current count: **zero**.

**4. Requirement traceability — mapped, but only half the job.** Every one of the 206 checklist items now has a status against the actual code: which of the 28 automated checks covers it, which are true by inspection but untested, which the agent deliberately does not attempt, and which don't apply because the design took a different path. See the two pages that follow. But be precise about what changed: this is a mapping *document*, built once and checked in alongside the code. It is not yet wired into `revalidate.py`'s own output — a green run still prints pass/fail, not requirement IDs. The gap the review named — “the release output should show, for every check, which requirement it covers” — is half-closed: the mapping exists, the automation to surface it live does not.

### The principle behind this page

Recording something as open until an automated check enforces it, rather than the moment prose claims it, is the right discipline — it is exactly what caught the fallback taxonomy and the paired-seed criterion earlier in this book. It should cut both ways, though: once a check exists and is green, leaving the item marked open costs the next boss real time re-verifying something already proven. This page exists so both states are visible at once, and so the next revision has to earn its own line rather than repeat this one.

## Where Every Checklist Item Actually Stands

206 items, one status each, checked against the real code rather than assumed from the checklist text. Five kinds of answer, and the difference between them is the whole point.

| Status                    | Count | What it means                                                                                                                                                                                                                                                                                    |
|---------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tested                    | 56    | A specific automated check verifies this — named in the evidence column of the companion spreadsheet. Where the measured result falls short of the checklist's own bar (two items do), the note says so plainly; “Tested” means instrumented and reported honestly, not that the target was hit. |
| Implemented               | 64    | True of the code, checked by reading it — but no dedicated automated check asserts it on every run. The largest single bucket, and an honest one: most of a working system is like this.                                                                                                         |
| Built, Disabled by Policy | 9     | Real, working code that the farm chooses not to run — town-hunger forecasting and the demand machinery beneath it. Measured, found harmful, switched off. Different from “not implemented” in every way that matters for a handover.                                                             |
| Not Implemented           | 42    | The capability genuinely does not exist — animals, fertilizer, telemetry logging, counterfactual branching, per-cause forecast-error classification. No code to point to.                                                                                                                        |
| N/A — different design    | 35    | The checklist item describes a mechanism the agent deliberately doesn't use — BFS pathfinding on a board with no obstacles, a separate planner module, oscillation detection for a re-planning loop that doesn't exist. Not a gap; a different route to the same place.                          |

Of the 108 **Critical**-priority items, 43 are formally Tested. The rest are split across the other four categories in roughly the same proportion as the full set — criticality tracked what mattered to verify, not what got verified.

### By section

| Section                | Items | Tested | Impl. | Disabled | Not built | N/A |
|------------------------|-------|--------|-------|----------|-----------|-----|
| Runtime & Game Control | 10    | 7      | 3     | 0        | 0         | 0   |
| State Management       | 8     | 0      | 5     | 0        | 0         | 3   |
| Movement & Pathfinding | 8     | 1      | 5     | 0        | 0         | 2   |
| Plant Lifecycle        | 7     | 2      | 4     | 0        | 0         | 1   |
| Watering & Daily Care  | 7     | 3      | 4     | 0        | 0         | 0   |
| Crops & Production     | 8     | 2      | 4     | 0        | 0         | 2   |
| Fertilizer             | 5     | 0      | 0     | 0        | 5         | 0   |
| Animals                | 11    | 0      | 0     | 0        | 11        | 0   |
| Labor                  | 7     | 2      | 2     | 0        | 0         | 3   |
| Land                   | 7     | 0      | 5     | 0        | 0         | 2   |
| Warehouse / Shed       | 9     | 2      | 4     | 0        | 2         | 1   |
| Market & Orders        | 11    | 4      | 3     | 3        | 0         | 1   |
| Demand Forecast        | 6     | 0      | 0     | 6        | 0         | 0   |

| Section                     | Items | Tested | Impl. | Disabled | Not built | N/A |
|-----------------------------|-------|--------|-------|----------|-----------|-----|
| Forecasts                   | 5     | 0      | 2     | 0        | 3         | 0   |
| Terminal Forecast           | 7     | 1      | 4     | 0        | 0         | 2   |
| Replanning / Counterfactual | 10    | 0      | 1     | 0        | 7         | 2   |
| Task Scheduling             | 10    | 4      | 5     | 0        | 0         | 1   |
| Validation                  | 8     | 6      | 1     | 0        | 0         | 1   |
| Endgame                     | 9     | 2      | 6     | 0        | 0         | 1   |
| Telemetry                   | 10    | 0      | 0     | 0        | 10        | 0   |
| Forecast Error              | 6     | 0      | 1     | 0        | 2         | 3   |
| Performance                 | 5     | 1      | 2     | 0        | 0         | 2   |
| Robustness                  | 10    | 5      | 0     | 0        | 2         | 3   |
| Evaluation                  | 8     | 6      | 0     | 0        | 0         | 2   |
| Architecture & Release      | 14    | 8      | 3     | 0        | 0         | 3   |

Three sections are worth reading before the rest: **08 Animals** and **07 Fertilizer** are 100% Not Implemented, which matches everything else this book says about them. **13 Demand Forecast** is 100% Built-Disabled — every item in that section has real code behind it, all of it switched off by the same measured result. **20 Telemetry** is 100% Not Implemented and is the plainest reading of Handover section 9's “still only intention.”



## A Sample of the Mapping

One row from each status, so the categories mean something concrete rather than staying abstract. All 206 rows, with full evidence for every one, ship as a companion file: `requirement_traceability.csv`.

| ID  | Item                                | Status                    | Evidence                                                                                                                                                                                                         |
|-----|-------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 097 | MAX_ORDERS enforced                 | Tested                    | <code>compliance_audit.py</code> : ‘market orders exceed cap’ — 0 violations across 62,545 turns.                                                                                                                |
| 045 | Crop EV per tile calculated         | Implemented               | <code>crop_values()</code> computes $(\text{units} \times \text{price} - \text{seed}) / \text{cycle}$ per crop every turn. True by reading the code; no dedicated check asserts the arithmetic on every run.     |
| 099 | Town consumption schedule modeled   | Built, Disabled by Policy | <code>drain_per_day()</code> reads live <code>town_model</code> data correctly. <code>policy.demand_forecast=False</code> : measured $-\$4,037$ to $-\$8,900/\text{season}$ against two opponents, switched off. |
| 056 | Animal purchase generated correctly | Not Implemented           | No <code>BUY_ANIMAL</code> is ever placed. <code>ANIMALS_ENABLED</code> defaults false and is checked in the scheduler itself, not merely tested for — see Revision 5, item 2.                                   |
| 023 | BFS returns executable first step   | N/A — different design    | No BFS exists. The board has no obstacles — locked tiles are walkable — so greedy Manhattan-distance stepping is provably optimal and was chosen deliberately.                                                   |

**What this does and doesn't prove.** Every classification above came from reading `agent_rd.py`, `revalidate.py` and `compliance_audit.py` directly, not from the checklist's own wording — several items turned out to describe mechanisms (BFS, a separate planner module, oscillation detection) the agent never intended to build, and those are marked N/A rather than forced into Tested or Not Implemented. This is a snapshot from one careful pass, not a self-updating gate: change the code without re-running `build_traceability.py` and this table will quietly go stale exactly the way the Monte Carlo numbers do. Treat it the same way — current as of its last generation, not as of the last rebuild.